

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

In Ignorance, find out how we store and retrieve memories in our brains. Then, head on over to Space Age to learn more about the local, star creating gas bubble that our Solar System is now inside.



Strong guts

A study funded by the National Science Foundation has found out that certain microbes living in the guts of squirrels helps them stay strong during their hibernation period. These microbes help the squirrels extract the remaining nitrogen from residual area to convert them into amino acids, which are in-turn used to preserve muscles.

<https://digit.in/strongguts>

WHAT'S NEW

Master of the waters, now driving on land

A series of experiments conducted by researchers from the Ben-Gurion University of the Negev, Israel, found that there was a possibility of harnessing the ability of fishes to navigate underwater to make them drive vehicles on land. They used a trained goldfish and a specially developed Fish Operated Vehicle or FOV.

During the tests, the researchers placed the fish in a water tank mounted on top of the FOV and positioned it such that it was able to see food placed at a distance. Fitted with a LIDAR sensor, a camera, and a computing unit, the FOV was programmed to react "to the fish's movement characteristics, location and orientation in its water tank to change the



vehicle's; i.e., the water tank's, position in the arena."

The fish in the tank was able to manoeuvre the FOV towards the

bait, avoiding dead ends and all the obstacles that came its way. Multiple rounds of testing suggested the same thing. The researchers thus deduced that the fish was capable of translating its ability to recognise the space around it and navigate towards a defined target when moved into a different terrestrial environment.

<https://digit.in/fishdriver>

Making disinfectants from sawdust

A team of researchers from Fudan University, Chongqing University, Shanghai Institute of Pollution Control and Ecological Security, and the Chinese Academy of Medical Sciences, have published a paper titled "Biosafety of human environments can be supported by effective use of renewable biomass". They showed that

certain forms of biomass could be converted into anti-microbial substances that are not only capable of neutralising several pandemic-threatening microbes but are also environmentally friendly.

The team used the process of hydro-thermal liquefaction to extract bioliquid from the given biomass. The fluid was deemed to

be safe for use after it went through rabbit skin testing.

According to the team of researchers, the cost of producing this newly found disinfectant was \$0.04427 per kilogram, which was deemed to be significantly cheaper than that of the existing chemical alternatives. The disinfectant, could play a pivotal role in reducing the negative impact of the use of the current alternatives.

<https://digit.in/biosafety>



Arithmetic of smell

Researchers at the McKelvey School of Engineering at Washington University were able to devise an algorithm capable of predicting whether the test subject, a locust, was able to sense an odour or not. The algorithm accurately predicted whether or not the locust would react to an odour.

<https://digit.in/smellmath>



Saving sharks and turtles

A research study published recently has found that installation of green LED lights on gillnets during fishing could prevent the entangling of a lot of "non-target" organisms, which include a variety of biologically important and endangered species.

<https://digit.in/ledfishing>



Salival relations

In a research paper titled, "Kids attend to saliva sharing to infer social relationships", it has been found that kids tend to form intimate relationships with individuals whom they exchanged saliva with. The children used the saliva exchange as cue to form "thick" relationships.

<https://digit.in/thickre>